Mustafa Alhelawe

NYC Metro · Open to Relocation | (732) 337-8562 | mustafa.alhelawe@gmail.com | mustafa-alhe.dev

SKILLS

Languages: Scala, JavaScript, TypeScript, Python, C#, Java, SQL, SPARQL, Bash
Backend & Architecture: Distributed Systems, Microservices, Event-Driven Architecture, REST APIs (Scalatra), RxScala, Node.js, PostgreSQL
Messaging & Protocols: FIX Protocol (QuickFIX/J), Pub/Sub Messaging (AMPS), Protobuf, Circe/JSON, DTCC CTM
Frontend: React, Redux, Next.js, React Native, Expo
Testing: ScalaTest, Mockito, Cucumber, Gatling, Cypress, Jest
Tools: CI/CD (Jenkins), Gradle, Ansible, Docker, Linux, Git, Splunk, Unity, Claude, Codex

WORK EXPERIENCE

Software Engineer — **Jersey City, New Jersey**
Bank of America Merrill Lynch — January 2023 – Present

- Shipped features across 10+ trade flows (matching, booking, amendment, cancellation) in Scala services for a client-facing, event-driven post-trade allocation and confirmation platform processing 60K+ institutional allocations daily.
- Extended services linking FIX allocations to 200K+ daily orders from multiple upstream systems over AMPS pub/sub and file-based feeds, enabling real-time matching and DTCC CTM affirmation.
- Built per-client, per-currency gross and commission tolerances into the linking engine via a config-first fallback chain, shipped behind a feature flag with a versioned event schema for safe replay.
- Built Python CLI tools and a Jenkins MCP server giving AI agents build-log, PR, and Jira context; also use Python for production log analysis, AMPS monitoring, and Ansible template validation.
- Onboarded Mexico flow for the Mexico T+1 initiative, building the filter that ingests Mexican upstream orders and replacing a hardcoded booking special case with a general fallback chain so pre-allocated allocations book.
- Designed and led self-service client routing: onboarding/amendment portals, APIs, and maker-checker approvals in the exception queue, replacing biweekly per-client code releases; migrated existing configs to SOW.
- Built bulk approve/reject for the operations exception queue (React/Redux, REST APIs) with partial-failure handling, letting operations teams clear trade mismatches and routing/config approvals in batches instead of one at a time.
- Delivered a client-tier attribute end-to-end across five microservices (SPARQL reference-data queries, Protobuf/JSON contracts, React grids), letting operations teams prioritize exceptions for top-tier clients.
- Led a zero-downtime scale-out from 8 to 16 production servers and procured a new 8-server preprod environment, maintaining Ansible and Autosys deployment configuration across all environments.

Cybersecurity Analyst Intern — **East Rutherford, New Jersey**
Capri Holdings Limited — June 2022 – August 2022

- Triaged LogRhythm security alerts, analyzed suspicious emails in CrowdStrike's sandbox, and assessed vendor security risks.

PROJECTS

Splits — Automated Shared Expense Tracking | splitshq.com — February 2026 – Present

- Built an expense-sharing application with a Next.js web app and React Native/Expo mobile clients, supporting recurring bills, customizable expense splits, and balance tracking.
- Detected recurring bills from Plaid transactions using normalized merchant names, amount clustering within a 15% tolerance, and recurrence across 2+ months; generated suggested split rules for automatic shared-bill entries.
- Built incremental Plaid transaction syncing with per-item concurrency locks and recovery from pagination errors, restarting paginated retrieval when transaction data changed during a sync.
- Applied generated bill entries before advancing the transaction-sync cursor, using database-enforced idempotency keys to prevent duplicate ledger entries when a sync is retried or replayed.
- Moved the mobile backend into Next.js API routes so web and React Native/Expo clients share Supabase authentication and PostgreSQL storage, eliminating a separate backend deployment.

Reinforcement Learning Simulations — January 2025 – Present

- Built a series of reinforcement-learning simulations in Unity ML-Agents, implementing modular C# environment logic and Python training pipelines to train autonomous agents using PPO.
- Progressed from food-seeking without vision to navigation around static and moving obstacles, then visual observations, refining observation design and reward functions through successive training experiments.
- Extended reinforcement-learning experiments to a 6-axis robotic arm, applying the same simulation approach to more complex observations, action spaces, reward functions, and robotic control.

EDUCATION

Rutgers, The State University of New Jersey — **New Brunswick, New Jersey**
Bachelor of Science in Electrical and Computer Engineering — Class of 2022